

SEMESTER END EXAMINATIONS – APRIL 2023

Program	: B.E. - CSE (Cyber Security) / CSE (Artificial Intelligence & Machine Learning)	Semester	: III
Course Name	: Data Base Management Systems (Integrated)	Max. Marks	: 100
Course Code	: CY32/CI32	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Identify the role of Database Administrator (DBA) and Database Designers. Discuss in detail. CO1 (06)
 - Classify any three Actors on the scene. Explain each with their responsibilities. CO1 (06)
 - Draw ER Diagram for each of the following Statements with (min,max) notation. Identify and mention relevant attributes for each. CO1 (08)
 - A man can be married to at most one women and vice versa.
 - An airport lies in exactly one country. A country may have arbitrarily many airports(and maybe none at all)
 - One instructor teaches many courses, but each course is run by exactly one instructor.
 - Each department has exactly one department head, some employees are the head of one department.
- Discuss any three advantages of DBMS Approach. CO1 (06)
 - With a neat diagram, explain the following with respect to database system environment. CO1 (07)
 - User
 - Interface
 - Database Languages.
 - Design an ER diagram by considering the scenario given below CO1 (07)
 - A salesperson may manage many other salespeople. A salesperson is managed by only one salespeople. A salesperson can be an agent for many customers.
 - A customer is managed by one salespeople. A customer can place many orders. An order can be placed by one customer. An order lists many inventory items.
 - An inventory item may be listed on many orders. An inventory item is assembled from many parts.
 - A part may be assembled into many inventory items. Many employees assemble an inventory item from many parts. A supplier supplies many parts. A part may be supplied by many suppliers.

Identify each entity with appropriate set of attributes. Construct ER diagram by specifying the structural constraints.

UNIT – II

- Discuss the Constraint violations happens during the update operation. Illustrate with examples. CO2 (06)

- b) Design the Relational Schema for the Entity-Relationship Diagram as CO1 (08)
shown in Fig.3(b) based on the ER to Relational Mapping Algorithm.

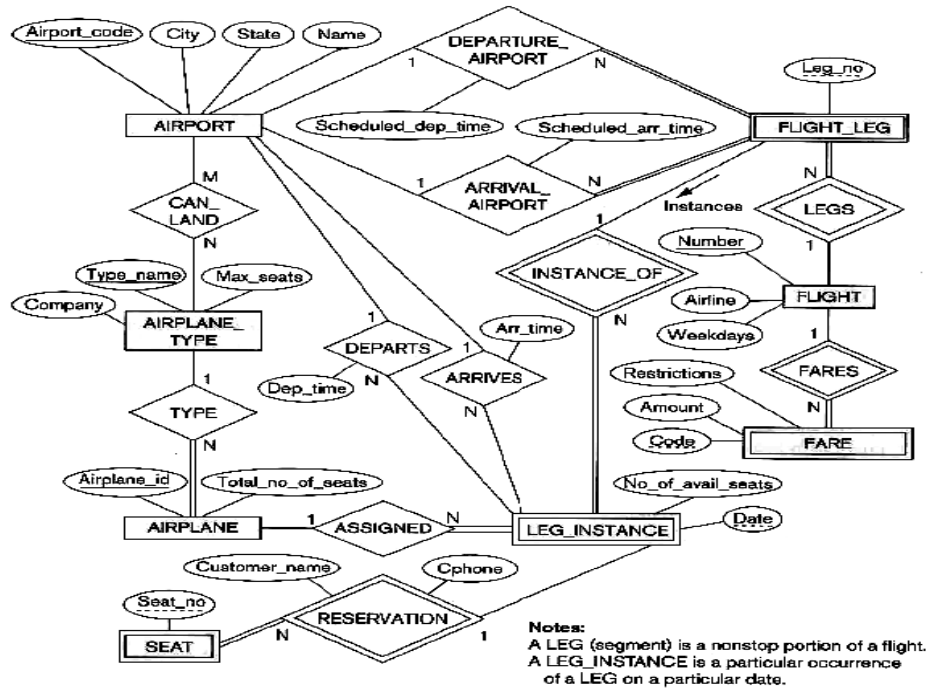


Fig.3(b) ER diagram

- c) Consider the schema given below. CO2 (06)

Student (USN, Name, Branch, cgpa)
Faculty (F_ID, Name, Department, Designation, salary)
Course (C_ID, Name, F_ID)
Enroll (C_ID, USN, Grade)

Give the relation algebra expression for the following statements

- List the USN and names of students who have secured 'B' grade in 'CI02' course.
- List the names and salaries of faculties who are professors.
- For each department, find the maximum salary of instructors in that department.

4. a) Differentiate between a key and a candidate key? Explain with an example. CO2 (07)

- b) Consider the following Supplier part relational schema. CO2 (06)

Suppliers (sid : integer, name : string, address : string)
Parts (pid : integer, pname : string, color : string)
Supply (sid : integer, pid : integer, cost : real)

Write relational algebra queries for the following.

- Find the sids of suppliers who supply red or green parts.
- Find the sids of suppliers who supply every part.
- Find the sids of suppliers who supply every red part.

- c) Consider the following relation schema CO2 (07)

Instructor(ID, name, dept_name, salary)
Teaches(ID, course_id, sec_id, semester, year)
Course(course_id, title, dept_name, credits)

- Write relational algebra query for finding the names of all instructors in the Music department who have taught a course in 2019, along with the titles of the courses they taught.
- Discuss whether this delete operation violates any integrity constraints.
DELETE from Course where dept_name='Sports';

UNIT – III

5. a) What do you mean by nested queries in SQL? Why IN operator is preferred over JOIN in nested queries? Explain with an example. CO3 (06)
- b) Consider the following schema. CO3 (08)
- SAILORS(SID, SNAME, RATING, AGE)
- BOATS(BID, BNAME, BCOLOR)
- RESERVES(SID, BID, DAY)
- Formulate the SQL queries for the following.
- Create the above tables by imposing the following constraints
Rating can be an integer between 1 and 10
 - Find the names of sailors who have reserved Yellow boat but not Orange boat
 - Find the names of sailors who have not reserved Interlake boats
 - For each red boat find the number of reservations made.
- c) Differentiate between JOIN ON and NATURAL JOIN operations in SQL. CO3 (06)
6. a) Illustrate how are views created and deleted in SQL. Give examples. CO3 (06)
- b) Consider the schema given below: CO3 (08)
- EMP(Ssn, fname, lname, bdate, address, sex, salary, superssn, dno)
- DEPT(dnumber, dname, mgrssn, mgrstartdate)
- PROJECT(pnumber, pname, ploc, dnum)
- WORKS_ON(essn, pno, hours)
- DEPENDENT(Dname, Essn)
- Mention SQL query for the following:
- To change the location and controlling department number of project number 10 to "Bellaire" & 5 respectively.
 - Retrieve all employees in department 5 whose salary is between \$30,000 & \$40,000.
 - Retrieve a list of employees & the projects they are working on, ordered by department & with in each department ordered alphabetically by last name, first name.
 - Retrieve name of employees with no dependents.
- c) Consider the Supply-Parts database given below. Assume relevant data types for attributes. CO3 (06)
- SUPPLIER (Sid, Sname, Address)
- PART (PID, Pname, Color)
- SHIPMENT (Sid, PID, Cost)
- Mention SQL query for the following
- Find the Sid's of suppliers who supply a yellow part.
 - For every supplier print the name of the supplier and the total number of parts that he/she supplies.
 - Update the part color supplied by supplier s3 to green.

UNIT- IV

7. a) Given the relation r(R). State whether the following functional dependencies are satisfied by the relation or not. CO4 (06)
- i) $A \rightarrow B$ ii) $AB \rightarrow C$ iii) $C \rightarrow A$ iv) $BC \rightarrow A$

A	B	C
1	4	2
3	4	6
3	4	6
7	3	8
9	1	0

- b) A set of FD's of a relation $R(W,X,Y,Z)$ is $\{X \rightarrow YZ, Y \rightarrow Z, X \rightarrow Y, XY \rightarrow Z, XZ \rightarrow W\}$. Find the minimal cover for this set of Functional Dependencies. CO4 (08)
- c) Let $R = ABCDE$, $R_1 = AD$, $R_2 = AB$, $R_3 = BE$, $R_4 = CDE$, and $R_5 = AE$. Let the functional dependencies be: $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $DE \rightarrow C$, $CE \rightarrow A$. CO4 (06)
Apply algorithm to test if the decomposition of R into $\{R_1, R_2, R_3, R_4, R_5\}$ is a lossless join decomposition.
8. a) Write an algorithm to find the closure of given functional dependency. CO4 (09)
Given the following set of functional dependency check whether they are equivalent or not.
Relation $R = (A, B, C, D, E, F)$
°F 1 = $\{ A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow F \}$
°F 2 = $\{ A \rightarrow CD, E \rightarrow AF \}$
- b) Explain minimal cover of set of functional dependency. Write an algorithm to find the minimal cover set of the functional dependency. CO4 (05)
- c) Consider the relation R with four attributes A, B, C, D with the following sets of FD's CO4 (06)
 $C \rightarrow D, C \rightarrow A, B \rightarrow C$
i. Identify the candidate key(s) for R .
ii. Identify the best normal form that R satisfies (1NF, 2NF, 3NF, or BCNF).
iii. If R is not in BCNF, decompose it into a set of BCNF relations.

UNIT - V

9. a) Illustrate ACID Properties in Transaction. Discuss the isolation levels of transaction. CO5 (08)
- b) Consider a database with objects X and Y and assume that there are two transactions T_1 and T_2 . CO5 (06)
Transaction T_1 reads objects X and Y and then writes object X .
Transaction T_2 reads objects X and Y and then writes objects X and Y .
i. Give an example schedule with actions of transactions T_1 and T_2 on objects X and Y that results in a write-read conflict.
ii. Give an example schedule with actions of transactions T_1 and T_2 on objects X and Y that results in a read-write conflict.
iii. Give an example schedule with actions of transactions T_1 and T_2 on objects X and Y that results in a write-write conflict.
- c) Discuss the actions taken by the read item and write item operations on a database. CO5 (06)
10. a) Why concurrency control is needed? Discuss the types of problems that might be encountered if transactions are allowed to run concurrently. CO5 (09)
- b) Which of the following schedules is (conflict) serializable? For each serializable schedule, determine the equivalent serial schedules. Draw the precedence graph for each. CO5 (06)
i. $r_1(x); r_2(z); r_1(x); r_3(x); r_3(y); w_1(x); w_3(y); r_2(y); w_2(z); w_2(y)$
ii. $r_1(x); r_2(z); r_3(x); r_1(z); r_2(y); r_3(y); w_1(x); w_2(z); w_3(y); w_2(y)$
- c) Write the procedure for recovery based on immediate update in multiple user environment. CO5 (05)
